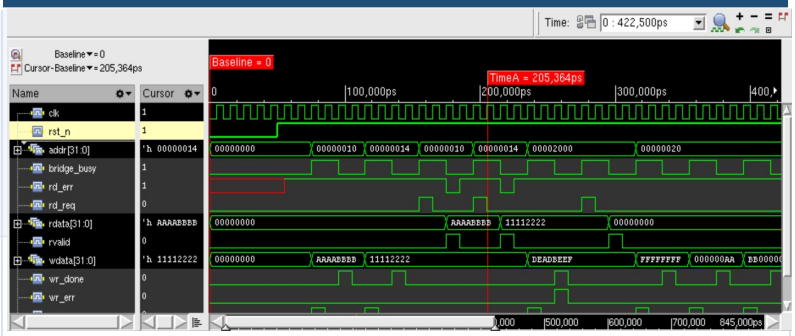
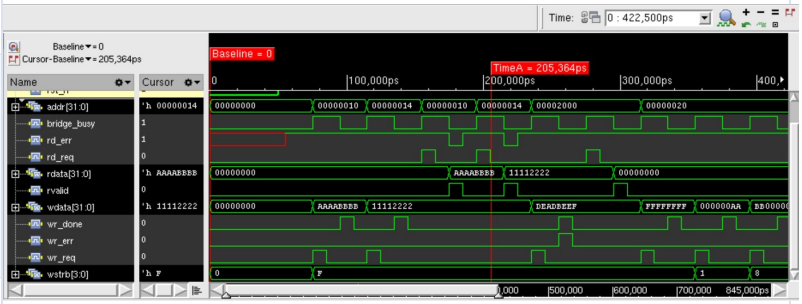
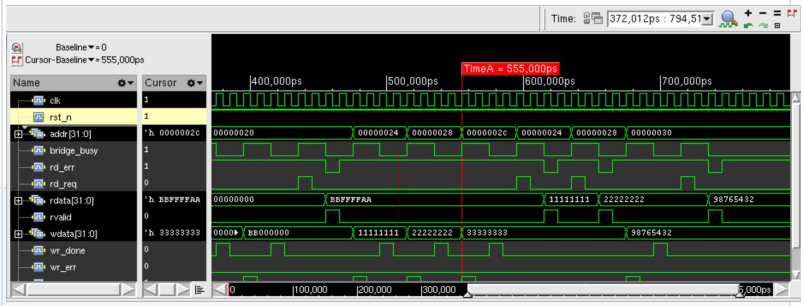
Test / Scenario	Stimulus / Method	Expected Result / Reference	Evidence
[TC1] Basic function: Valid 32-bit Store/Load	Write 0xAAAA_BBBB to address 0x10 and 0x1111_2222 to address 0x14 using cpu_write; read both addresses using cpu_read.	Read 0x10 = 0xAAAA_BBBB and read 0x14 = 0x1111_2222; rd_err remains deasserted; both checks → PASS.	
[TC2] Error handling: Write outside the 1KB memory region	Write 0xDEAD_BEEF to out-of-range address 0x0000_2000 using cpu_write.	wr_err is asserted to indicate a write address error (DECERR); error flag detected → PASS.	
[TC3] Error handling: Read outside the 1KB memory region	Read from out-of-range address 0x0000_2000 using cpu_read.	rd_err is asserted to indicate a read address error (DECERR); error flag detected → PASS.	
[TC4] Corner case: Partial write and byte masking (WSTRB)	Write 0xFFFF_FFFF to address 0x20 with WSTRB=4'b1111; overwrite the lowest byte with 0xAA using WSTRB=4'b0001; overwrite the highest byte with 0xBB using WSTRB=4'b1000; read back address 0x20.	Final read data = 0xBBFF_FFAA, confirming that only the byte lanes selected by WSTRB are updated → PASS.	
[TC5] Stress: Sequential memory transactions	Perform three cpu_write calls for addresses 0x24, 0x28, and 0x2C with data 0x1111_1111, 0x2222_2222, and 0x3333_3333; then read back 0x24 and 0x28.	Read 0x24 = 0x1111_1111 and read 0x28 = 0x2222_2222. Address 0x2C is written but not read back. The task sequence alone does not prove zero idle cycles or continuous back-to-back AXI transfers.	
[TC6] Data integrity: Read-After-Write (RAW) to the same address	Write 0x9876_5432 to address 0x30 using cpu_write; then read address 0x30 using cpu_read.	Read data = 0x9876_5432, confirming readback of the value written to the same address → PASS.	